



National Institutes of Health
Bethesda, Maryland 20892
<http://www.nih.gov>

Date: February 06, 2026

From: Salman Ahmad, Quality Assurance Manager
Department of Laboratory Medicine

Through: Josh Cohen, Laboratory Manager
Department of Laboratory Medicine

Subject: **SOLE SOURCE JUSTIFICATION AND APPROVAL**

To: **NIH, CLINICAL CENTER, CONTRACTING OFFICER**

1. AGENCY IDENTIFICATION AND CONTRACTING ACTIVITY

DLM Point of Contact: Salman Ahmad 301-451-8311
Procurement of Service Support of Rees Scientific Electronic Temperature Monitoring System in DLM
National Institutes of Health (NIH) Clinical Center, Department of laboratory Medicine
9000 Rockville Pike, bld.10, Room 2C306
Bethesda, MD 20892
COR or AO's Name, contact information: Sacdia Hassan

Contracting Activity:
NIH Clinical Center
Office of Purchasing and Contracts
6707 Democracy Blvd
Bethesda, MD 20892

2. NATURE AND/OR DESCRIPTION OF THE ACQUISITION BEING APPROVED

This is sole-source procurement for Rees Scientific to provide a temperature monitoring system service support for the Department of Laboratory Medicine (DLM).

3. DESCRIPTION OF THE SUPPLIES AND SERVICES REQUIRED TO MEET THE AGENCY'S NEEDS

The system shall be specifically designed as an environmental monitoring system (EMS) capable of sensing environmental parameters such as temperature, humidity, differential pressure, etc. in real time and monitoring and alarming based on the real-time readings from its sensors.

The system shall be fully Windows® based taking advantage of Windows® superior event-driven programming, Windows® menus, Windows® dialog boxes, Windows® multi-tasking etc. A "Windows® Like System" or "Windows® Like User Interface" is not an acceptable equivalent.

The system shall offer SQL software suitable for use with SQL server software and Crystal reports. The system shall be fully capable of functioning properly on Windows® Server 2012/R2® or 2016. Server products. It shall be possible to run different Workstations on different Windows® platforms simultaneously so that different people in DLM can use the system on the PC that is already in their workstation.

The system must be capable of monitoring up to 128 wireless or hardwired sensors per node and "**must**" be capable of handling both wireless, Wi-Fi, and hardwired sensors (hybrid) in any number combination on a single system. Inputs

shall be universal - capable of any type of analog or switch sensor monitoring including but not limited to, temperature, differential pressure, pH, salinity, humidity, flow rate, flood, power failure, security breach, fluid level, etc.

Access to the system shall be controlled by user name and password. There shall be 8 levels of access permissions including permission to:

- a. Enable alarms
- b. Inhibit single alarms for a short time only.
- c. Inhibit single alarms for unlimited time.
- d. Inhibit multiple alarms.
- e. Program the Access Control System
- f. Program the Environmental System Inputs
- g. Program Environmental System Outputs
- h. Modify user names and passwords.

System shall be capable of answering the telephone and shall give the caller the complete current alarm status information, which telephone number has responded to alarm messages, if any, and current probe readings with its synthesized voice.

It shall be possible to interrogate individual inputs from the phone, from the node or from the PC to obtain readings and complete alarm status at any time. Any input can be inhibited remotely by phone, with the proper access code.

The system shall have customizable telephone lists with 100 separate lists of 10 telephone numbers each for a total of 1,000 potential numbers as a default option per node. Each input shall be capable of being assigned a different set of day/night phone lists. The system shall have a telephoning delay that is programmable. There may be up to 5000 names entered from which you may choose the list names.

System shall be capable of calling or e-mailing alarm conditions to users for each individual input.

System shall possess Database Watch alarm that will indicate if the system has lost communication with the database, also capable of alarming locally, calling out via telephone or e-mailing .

System must provide a “Buddy Watch” feature (each node watches and polices the other) feature for additional security. This is very important as DLM has two nodes. This will allow each system to have the ability to check the node health of each other on a continuous basis and alarm on the others behalf should a failure take place.

System shall possess MPX Com and Battery alarm that will indicate if one of MPX-RMT input remote panels loses communication to the node or battery, also capable of alarming locally, calling out via telephone, e-mailing or text messaging.

System must have independent programmable alarm delays on each input to provide false alarm prevention. The user shall be able to inhibit any alarm for a preset time period. The alarm will automatically re-enable after timing out.

There shall be an audit log that documents program changes, input calibrations, and access code changes. The log shows what the conditions were, what they were changed to, and who changed them. There shall be an automatic backup of the node program whenever a program change is made.

Individual sensors can be programmed to log at the Global logging rate or choose individual logging rates up to 1440 times per day.

The system shall include a watchdog timer to reboot the server in the event of any software/hardware problems, a dead system alarm to warn users should the hardware fail, and built-in web access software.

Hardwired inputs shall be connected to nodes through the use of Buffered MPX Remote panels. Communications with the panels shall be supervised, and a trouble alarm shall sound if interruption of such communications should occur.

The system must offer (2) levels of alarm notification. “Pre-alarm Alert” and “Out of Range” alarm. The pre-alarm alert must be capable of notification by audible, by remote audible/visual, and by email means.

System security must allow the option to use either monitoring system specific passwords, or the users' existing Windows passwords. Using Windows passwords eliminates the need for a separate password and accompanying password management SOP specific to the monitoring system.

The hardwired Buffered MPX remote panels must be capable of storing all recorded data at a logging rate of every 5 minutes up to 7 days in the event of node failure.

Wireless 2.4 GHz ZigBee and Wi-Fi transmitters must be capable of buffering 7 days of readings history recorded at 5 minute intervals. A completely functional (100% turnkey) EMS as described or implied in this specification. Design and selection of equipment shall be consistent with the requirements of long-term trouble-free operation. This EMS shall be qualified for continuous operation 24 hours a day, 365 days a year.

The EMS configuration, make and design of the equipment, and the arrangement of the equipment within the assemblies, shall be such as to afford maximum accessibility to all parts and shall be subject to the Owner's review and approval. Acceptable engineering standards shall be followed and equipment shall meet all codes pertaining to safety and insurance regulations.

The estimated value of this obligation will be **\$21,593.43**

4. IDENTIFICATION OF STATUTORY AUTHORITY PERMITTING OTHER THAN FULL AND OPEN COMPETITION.

✓ **FAR 13.106-1(b) (1):** For purchases not exceeding the simplified acquisition threshold (SAT), contracting officers may solicit from one source if the contracting officer determines that the circumstance of the contract action deem only one source reasonably available (e.g. urgency, inclusive licensing agreements, brand name, or industrial mobilization).

☐ **FAR 13.501 (a) (1) (ii) & 2(i):** For actions for commercial supplies or services greater than the SAT but not exceeding \$6.5M, the citation is: Test Program for commercial Items, Section 4202 of the Clinger-Cohen Act of 1996.

5. DEMONSTRATION THAT THE PROPOSED CONTRACTOR'S UNIQUE QUALIFICATIONS OR THE NATURE OF THE ACQUISITION REQUIRES USE OF THE AUTHORITY CITED

No other EMS vendor has ISO 9001 certification which meets temperature monitoring requirements of both current Good Manufacturing Practices (cGMP) and DLM's accrediting agency, the College of American Pathologists (CAP). DLM wants to only do business with vendors that are certified as ISO 9001 compliant, to establish continuous improvement in its ability to monitor and preserve optimal clinical and research products.

6. A DESCRIPTION OF EFFORTS MADE TO ENSURE THAT OFFERS ARE SOLICITED FROM AS MANY POTENTIAL SOURCES AS PRACTICABLE

DLM has explored EMS temperature monitoring systems such as Senso Scientific, MesaLabs, Isensix and Rees Scientific. Rees Scientific widely used throughout the NIH campus has been proven NOT to have any security vulnerabilities while operating at the NIH. The DLM Sterility Service has already procured the Rees Scientific EMS therefore it is in the best interest of the rest of the department to use the same system. As previously stated, Rees Scientific is also the only vendor ISO 9001:2015 certified.

7. DEMONSTRATION THAT THE ANTICIPATED COST WILL BE FAIR AND REASONABLE

I hereby certify that this justification is accurate and complete to the best of my knowledge and belief. Based on the foregoing, I approve this urgent and compelling justification provided that the services herein described have otherwise been authorized for acquisition.

8. DESCRIPTION OF MARKET RESEARCH CONDUCTED AND THE RESULTS OR STATEMENT OF THE REASON MARKET RESEARCH WAS NOT CONDUCTED

Market research has been conducted with other manufacturers such as Senso Scientific, Mesa Labs, and Isensix. No other vendors were found to satisfy all DLM required features. An essential service characteristic is for DLM to maintain critical research, and reagents at their assigned temperatures. It is also required by FDA cGMPs that certain research be able to document the proper storage of components stored at room temperature, room humidity, under refrigeration and in -20C and -80C freezers among other monitored parameters.

9. ANY OTHER FACTS SUPPORTING THE USE OF OTHER THAN FULL AND OPEN COMPETITION.

Due to the strict regulatory nature of products maintained by DLM, any company installing and maintaining the monitoring system must be ISO 9001:2015 certified and also be ISO 17025 accredited for compliance purposes.

10. LISTING OF SOURCES, IF ANY, THAT EXPRESSED, IN WRITING, AN INTEREST IN THE ACQUISITION.

See Section # 8

11. STATEMENT OF THE ACTIONS, IF ANY, THE AGENCY MAY TAKE TO REMOVE OR OVERCOME ANY BARRIERS TO COMPETITION BEFORE ANY SUBSEQUENT ACQUISITION FOR THE REQUIRED SUPPLIES OR SERVICES.

DLM provides laboratory testing for all Clinical Center inpatients and clinic outpatients. This translates to more than 2 million test results per year. It is critical that DLM acquires an EMS that meets stringent requirements to minimize the loss of crucial reagents, specimens, and research samples essential to optimal operation. **Rees Scientific is the only vendor which satisfies DLM requirements for:**

- Local service technicians on the NIH campus for prompt service calls, repairs, or training as required.
- The EMS **“must”** be able to accept a “hybrid” configuration of hardwired, wireless, and Wi-Fi sensors on the same system.
- All material furnished shall be new and of the latest design available from a single manufacturer who is engaged in the manufacture and sale of EMS.
- The manufacturer shall have an installed base of at least **(3)** similar systems of **(100 monitored points or more)** as a reference at the NIH.
- System must provide a *“Buddy Watch”* feature (each node watches and polices the other) feature for additional security.
- The system shall include a watchdog timer to reboot the server in the event of any software/hardware problems, a dead system alarm to warn users should the hardware fail, and built-in web access software.
- System security must allow the option to use either monitoring system specific passwords, or the users’ existing Windows passwords.

12. ADMINISTRATIVE OFFICER/PROGRAM OFFICIAL CERTIFICATION

“I hereby certify that the information contained in this justification is accurate and complete to the best of my knowledge and belief.”

Official	Name & Title	Signature & Date
Project Officer	Salman Ahmad Quality Assurance Manager DLM	SALMAN AHMAD -S Digitally signed by SALMAN AHMAD -S Date: 2026.02.06 12:28:28 -05'00'
Project Officer's Immediate Supervisor	Joshua Cohen Laboratory Manager DLM	JOSHUA E. COHEN -S Digitally signed by JOSHUA E. COHEN -S Date: 2026.02.06 14:02:15 -05'00'

13. CONTRACTING OFFICER'S CERTIFICATION

“I hereby certify that the information contained in this justification is accurate and complete to the best of my knowledge and belief.”

[NAME AND SIGNATURE]
Contracting Officer